Database Management Systems
(Common to CSE, IT, AIML & CSE (DS))

	L	T	P	C
Course Code: 2501IT05	**2**	**0**	**2**	**4**

Course Outcomes:

At the end of the course, student will be able to:

CO1: Describe the Fundamental concepts of DBMS.

CO2: Interpret relational database using SQL.

CO3: Make use of normalization techniques for database design.

CO4: Illustrate the mechanisms of transaction management.

CO5: Optimize database performance with advanced indexing, query optimization, and robust backup and recovery strategies.

Mapping of Course Outcomes with Program Outcomes:

CO/PO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2									
CO2	2	3	2								
CO3	3		2								
CO4		2	3								
CO5			3	3	2						

Mapping of Course Outcomes with Program Specific Outcomes:

CO/PSO	PSO1	PSO2
CO1	2	
CO2		3
CO3	3	
CO4		2
CO5	3	

UNIT – I

Introduction to DBMS: History and Architecture,Need and Purpose of DBMS, Applications of DBMS, Data and Metadata, Levels of abstraction,Three-Schema Architecture, structure of DBMS, Database Users & Roles of DBA, Basics of Parallel and Distributed Databases, Specialty Databases – Spatial, Temporal, Multimedia, NoSQL.

View of Data : Concepts of Schema, Data Models(Relational, ER, Object-based, Semi-structured), Steps in Database Design, Database system environment, Centralized and Client Server architecture for the database.

Practice

1. Familiarization with installation of DBMS(oracle 10g).
2. Draw the schema model for a University Database System by identifying the main entities using draw.io/diagrams.net or any similar tool.

UNIT – II

Introduction to the Relational Model: Structure of Relational Model, concepts of domain, attribute, tuple, relation,Database Schema and Schema Diagrams, Keys in Relational Model: Primary, Candidate, Alternate, Foreign, Importance and Handling of NULL Values, Relational Algebra: Fundamental Operations.

Basic SQL: Data types, table definitions (create, alter), DML operations (insert, delete, update), basic SQL querying (select, where clause, projection), arithmetic&logical operations, SQL functions, Introduction to GROUP BY and HAVING clauses.

Practice

1. Querying and modifying the database using Data Manipulation Language commands - select, insert, update, delete.
2. Implementation of SQL Functions –Use group-by and having clause.

- Find Customer Referee using DML Operations
  - https://leetcode.com/problems/find-customer-referee/description/?envType=study-plan-v2&envId=top-sql-50
- Average Time of Process per Machine using aggregate functions.
  - https://leetcode.com/problems/average-time-of-process-per-machine/description/?envType=study-plan-v2&envId=top-sql-50
- DML Operations on given schema
  - https://www.hackerrank.com/challenges/revising-the-select-query-2/problem?isFullScreen=true
- Aggregate Operators
  - https://www.hackerrank.com/challenges/earnings-of-
  - employees/problem?isFullScreen=true
- Basic Queries for salaries of employees
  - https://www.hackerrank.com/challenges
- /salary-of-employees/problem?isFullScreen=true

UNIT – III

ER Model – Introduction to entities, attributes, entity sets, relationships, relationship sets, constraints. Specialization, generalization, inheritance, subclasses and super classes using ER diagrams.

Advanced SQL – Creating tables with relationships, implementing key and integrity constraints, nested and correlated subqueries, grouping, aggregation, ordering, different types of joins (natural, equi, outer, left, right, inner), views (updatable and non-updatable), relational set operations.

Practice:

1. Perform join operations: natural, equi-join, outer join, left, right, inner; assess query performance.
2. Perform set operations: union, intersection, difference.
3. Implement nested and correlated subqueries.
4. Creating and querying views and materialized views.

- Queries using join operations Ex: Product Sales Analysis I
  - https://leetcode.com/problems/product-sales-analysis-i/description/?envType=study-plan-v2&envId=top-sql-50
- Queries using Nested queries. Ex: Employees Whose Manager Left the Company
  - https://leetcode.com/problems/employees-whose-manager-left-the-company/description/?envType=study-plan-v2&envId=top-sql-50
- Queries using join operations
  - https://leetcode.com/problems/primary-department-for-each-employee/description/?envType=study-plan-v2&envId=top-sql-50
- Queries challenge the pads problem
  - https://www.hackerrank.com/challenges/the-pads/problem?isFullScreen=true
- Queries type of triangle problem

- https://www.hackerrank.com/challenges/what-type-of-triangle/problem?isFullScreen=true
- Joins
  - https://www.hackerrank.com/challenges/the-report/problem?isFullScreen=true

UNIT – IV

Normalization: Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependency (1NF, 2NF and 3 NF), concept of surrogate key, Boyce-Codd normal form (BCNF), Lossless join and dependency preserving decomposition, Fourth normal form(4NF), Fifth Normal Form (5NF).

Transaction Concept: Transaction State, Implementation of Atomicity and Durability, Concurrent Executions, Serializability, Recoverability, Storage, Recovery and Atomicity, Recovery algorithm.

Practice

1. **Implement SQL queries on a normalized database schema** using the given university database structure. The schema consists of the following tables:

- **Students** (StudentID, StudentName, Major)
- **Courses** (CourseID, CourseName, Credits)
- **Enrollments** (StudentID, CourseID, EnrollmentDate)
- **Instructors** (InstructorID, InstructorName, Phone)
- **Course_Instructors** (CourseID, InstructorID)

2. (a) Implementation of Data Control Language commands – grant and revoke.

(b) Implementation of Transaction Control Language commands - commit, savepoint, and rollback.

- Understanding Transaction, Commit and Rollback
  - https://www.codechef.com/learn/course/sql/SQ00LS10A/problems/TRA01
- Queries on normalized database.
  - https://www.hackerrank.com/challenges/symmetric-pairs/problem?isFullScreen=true
- Queries on occupations
  - https://www.hackerrank.com/challenges/occupations/problem?isFullScreen=true

UNIT – V

Indexing Techniques: Introduction, B+ Trees: Search, Insert, delete algorithms, File Organization and Indexing, Cluster Indexes, Primary and Secondary Indexes, Index data Structures, Hash Based Indexing: Tree base Indexing, Comparison of File Organizations.

Database Tuning: Introduction, Database Performance Tuning, Query Optimization: Introduction, Query Optimization algorithms, Backup and Recovery.

Practice

1. Create a Primary, Secondary Index on a Column.
2. Retrieve Data Using an Index.
3. Insert Data and Update Indexes.
4. Delete Data and Impact on Indexes.

* **Note:** The student must Complete & Submit a Database Foundations Certificate Course offered by Oracle Academy at the end of the Practice Session.

Textbooks:

1 Database System Concepts ,Abraham Silberschatz, Henry Korth, and S. Sudarshan, McGraw-Hill , 8th Edition, ISBN-13: 978-1260230508.

2 Database Management Systems, Raghurama Krishnan, Johannes Gehrke, Mc Graw-Hill, 4th Edition, ISBN: 978-1260091195.

Reference Books:

1 Database Systems, Ramez Elmasri, Shamkant B. Navathe, 7th Edition, Pearson, ISBN: 978-0137504277.

2 Database Systems, Carlos Coronel, Steven Morris, Peter Rob, 10th Edition, Cengage, ISBN: 978-0357517864.

3 Introduction to Database Systems, C J Date, Pearson, 9th Edition, ISBN: 978-0133970777.

Web Links:

1 https://academy.oracle.com/pages/coursedescription/Oracle%20Academy%20Database%20Foundations%20Course%20Description.pdf

2 https://www.w3schools.com/sql/

3 https://onlinecourses.nptel.ac.in/noc22_cs91/

Additional Practice:

S.No	Difficulty	Problem Name	Link
1.	Easy	Select	https://www.codechef.com/learn/course/sql/SQ00LS04/problems/GSQ17
2.	Easy	DML queries	https://leetcode.com/problems/second-highest-salary/description/
3.	Easy	Select All	https://www.hackerrank.com/challenges/select-all-sql/problem?isFullScreen=true
4.	Easy	Basic employee names	https://www.hackerrank.com/challenges/name-of-employees/problem?isFullScreen=true
5.	Easy	Aggregate functions	https://www.hackerrank.com/challenges/revising-aggregations-sum/problem?isFullScreen=true
6.	Easy	Basic SQL Querying	https://www.codechef.com/learn/course/sql/SQ00LS04/problems/GSQ21
7.	Easy	Basic Join	https://www.hackerrank.com/challenges/the-report/problem?isFullScreen=true